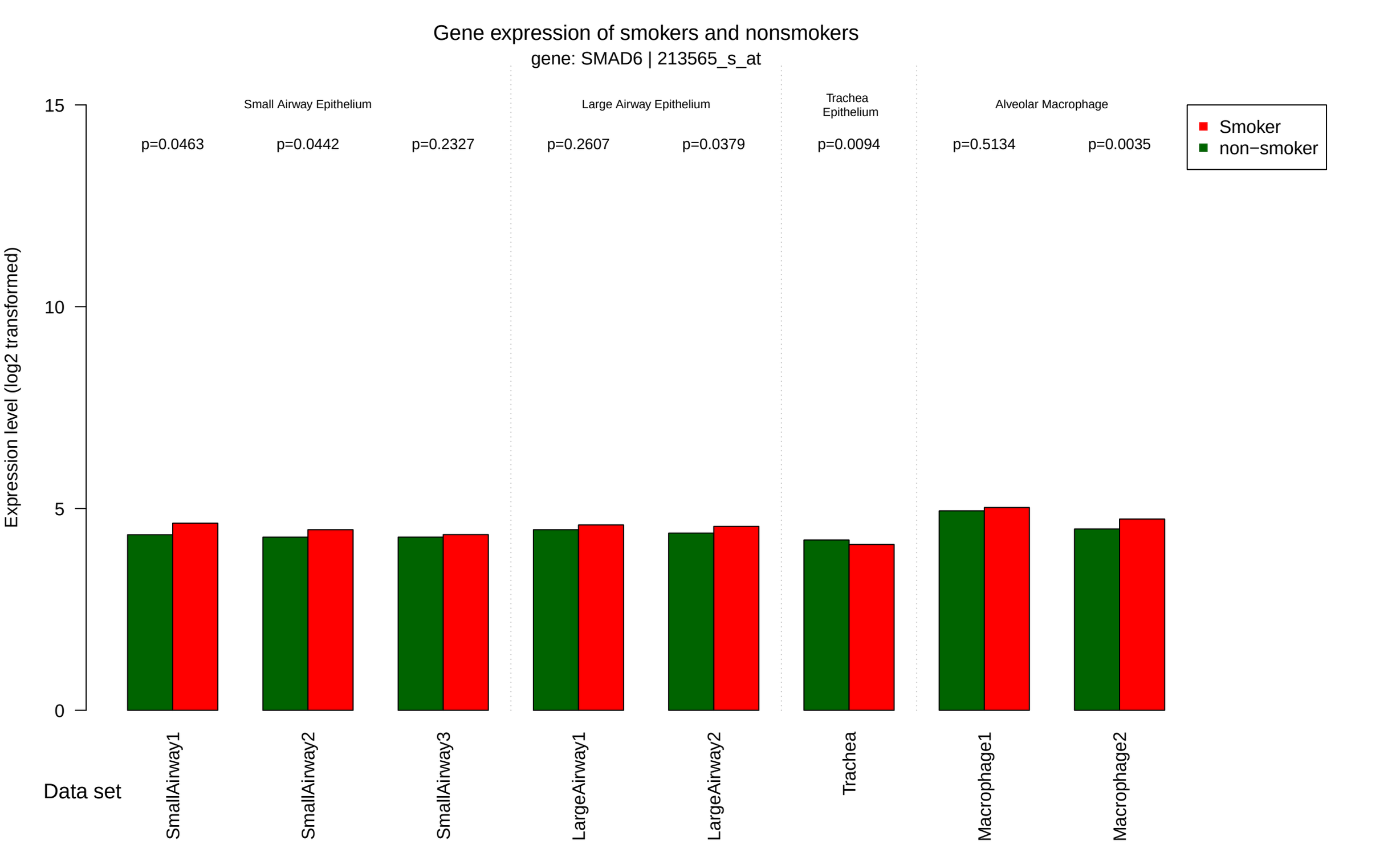
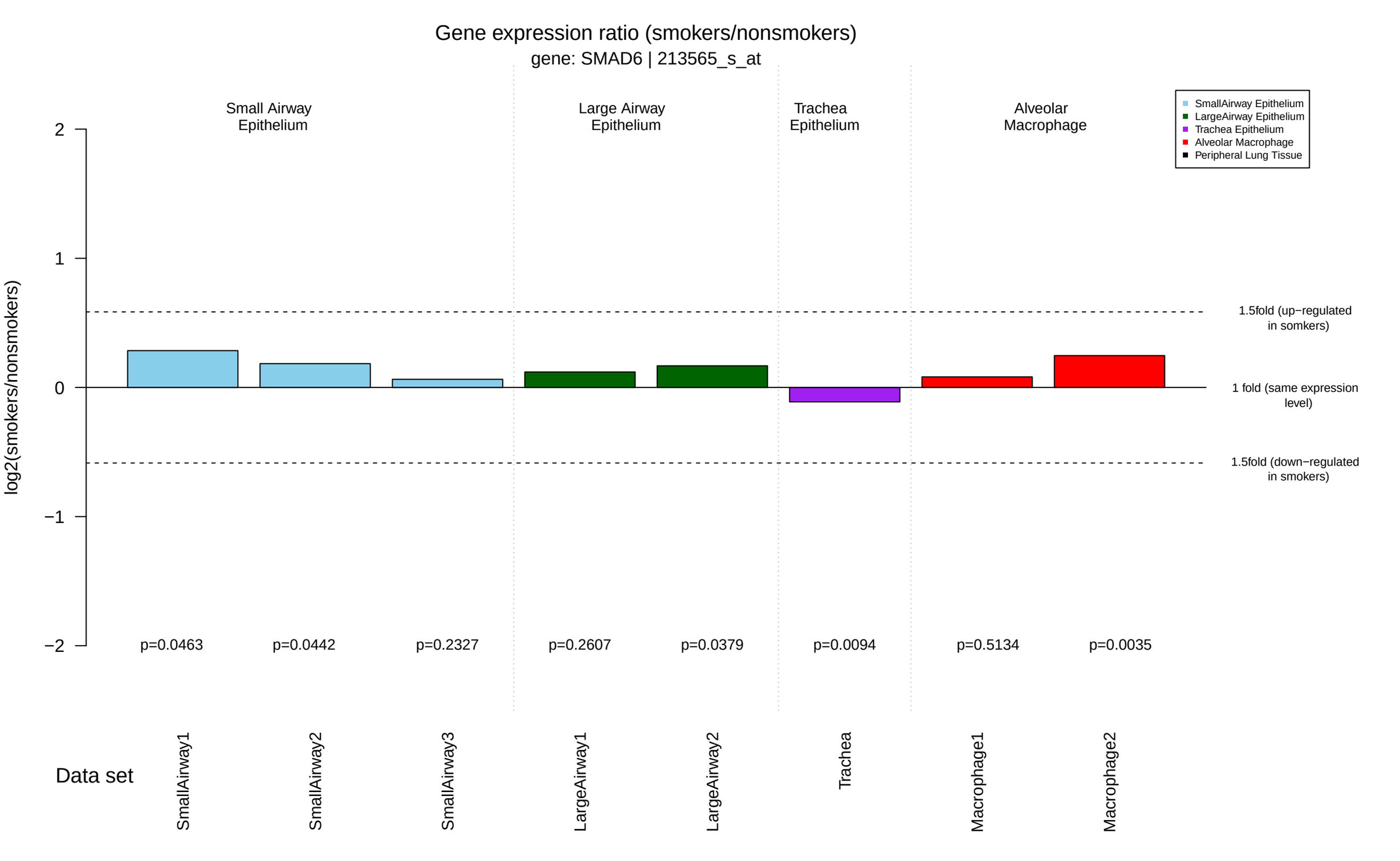
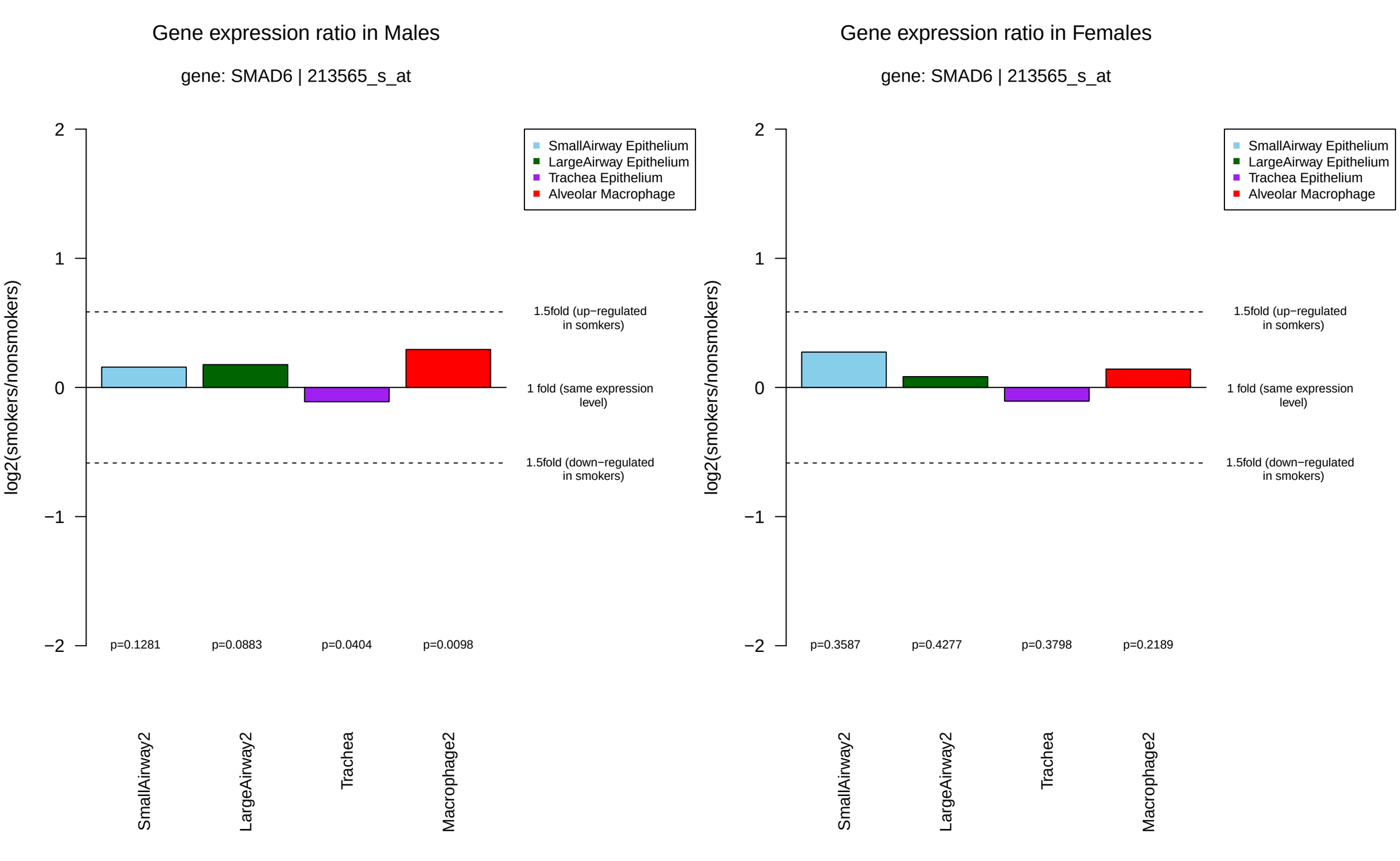
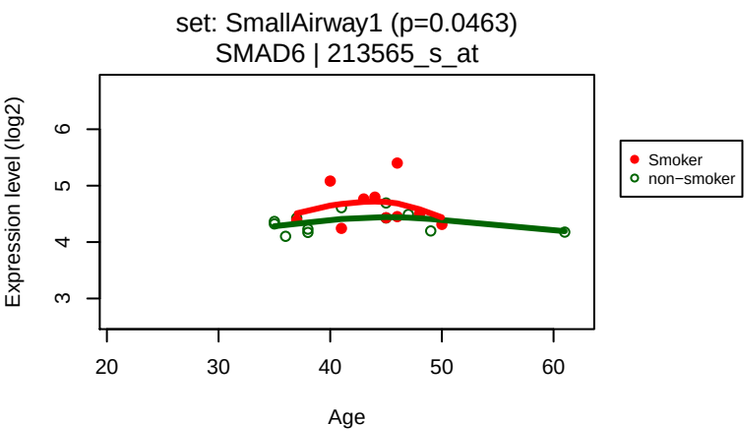
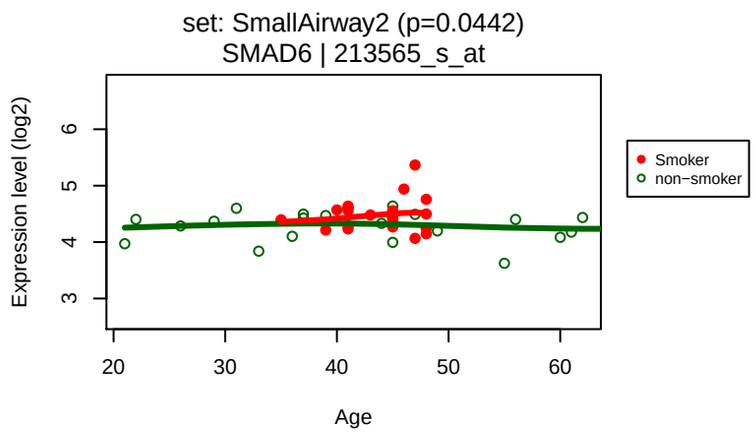
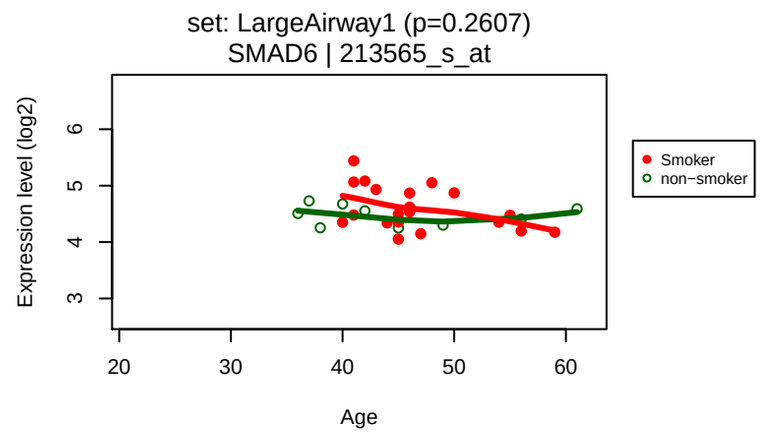

Gene expression of smokers and nonsmokers
gene: SMAD6 | 213565_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0463
p=0.0442
p=0.2327
p=0.2607
p=0.0379
p=0.0094
p=0.5134
p=0.0035
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

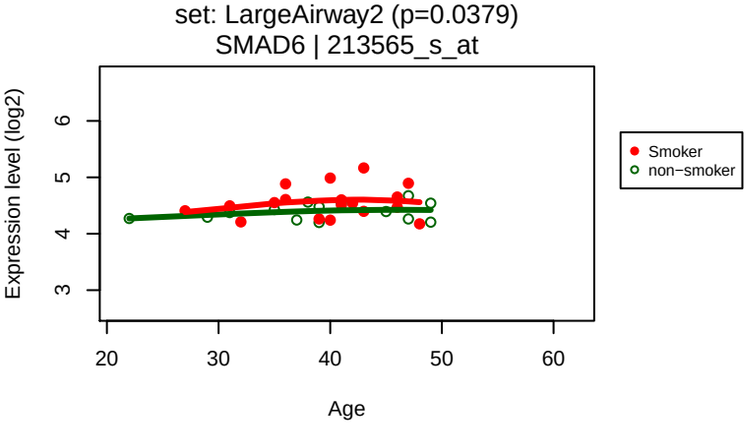

Gene expression ratio (smokers/nonsmokers)
gene: SMAD6 | 213565_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0463
p=0.0442
p=0.2327
p=0.2607
p=0.0379
p=0.0094
p=0.5134
p=0.0035
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

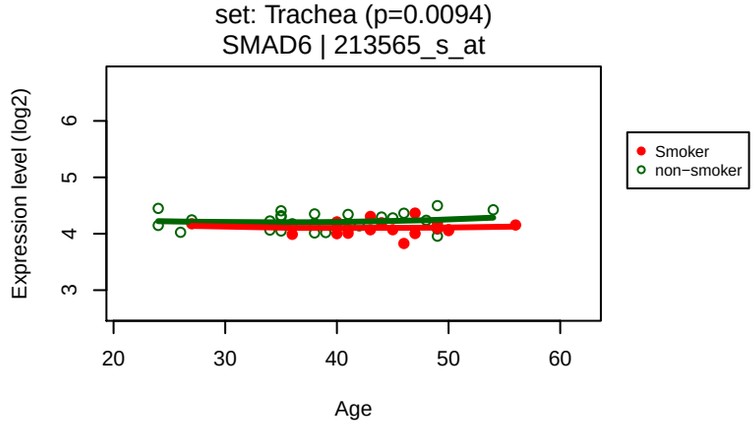

Gene expression ratio in Males
gene: SMAD6 | 213565_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1281
p=0.0883
p=0.0404
p=0.0098
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD6 | 213565_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3587
p=0.4277
p=0.3798
p=0.2189
SmallAirway2
LargeAirway2
Trachea
Macrophage2

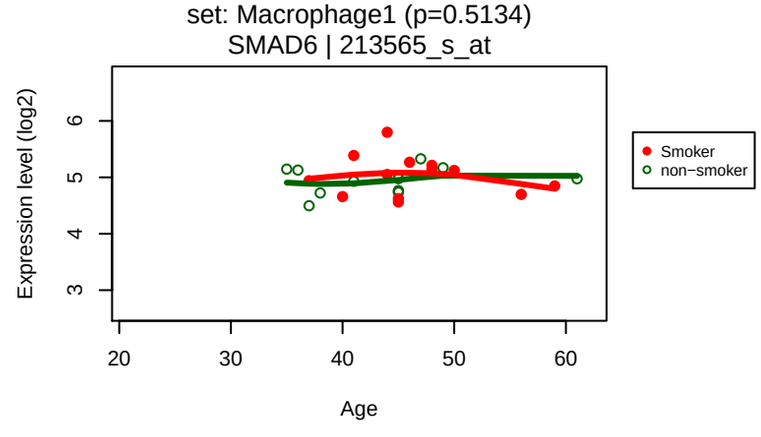

set: SmallAirway1 (p=0.0463)
SMAD6 | 213565_s_at
Expression level (log2)
Age
Smoker
non−smoker

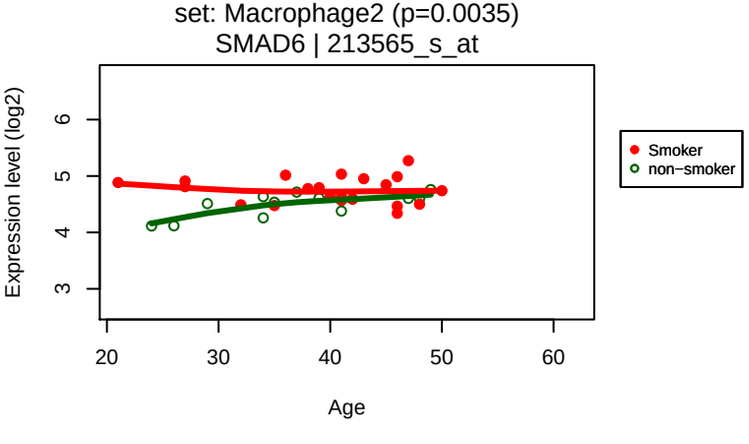

set: SmallAirway2 (p=0.0442)
SMAD6 | 213565_s_at
Expression level (log2)
Age
Smoker
non−smoker

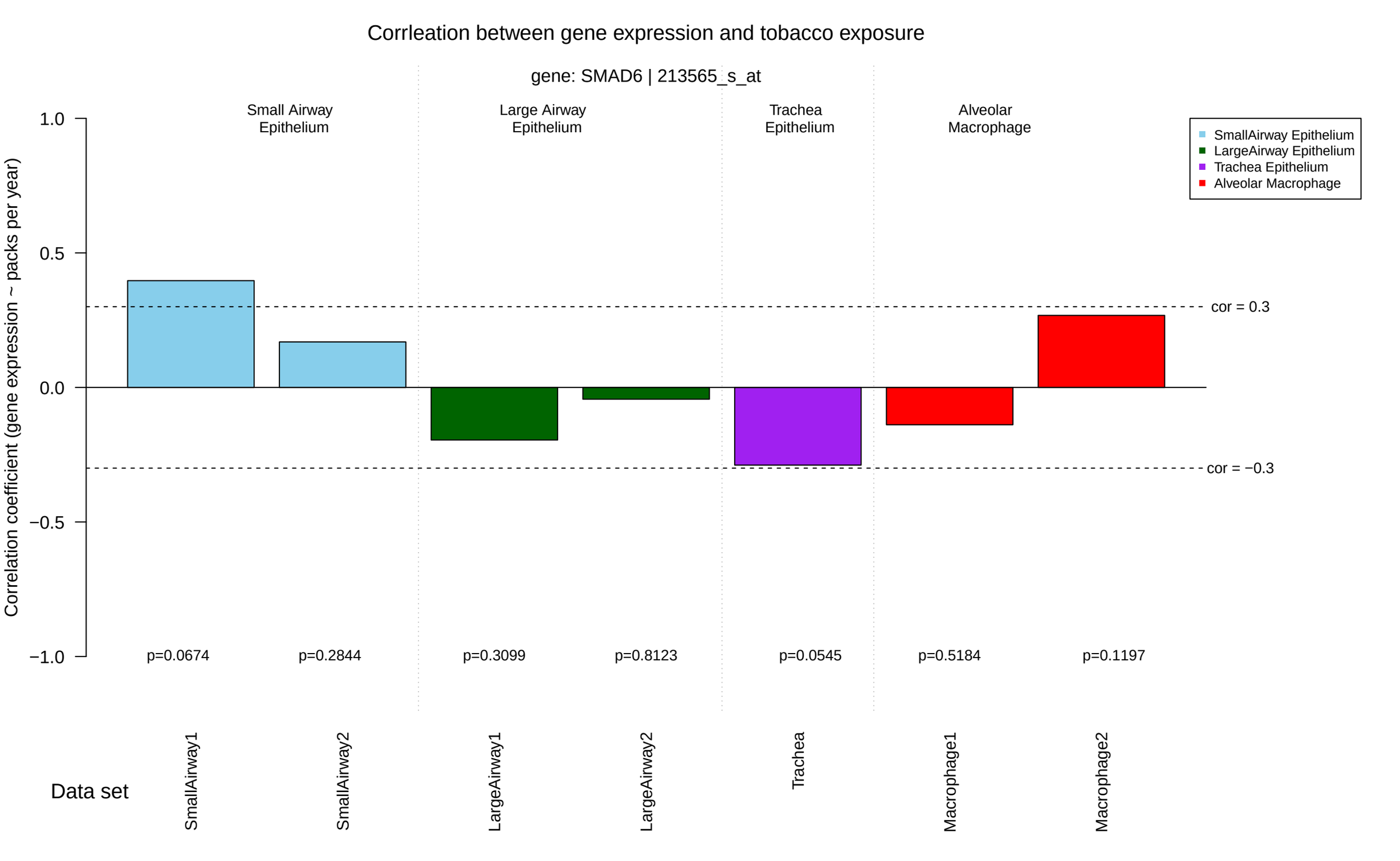

set: LargeAirway1 (p=0.2607)
SMAD6 | 213565_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0379)
SMAD6 | 213565_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0094)
SMAD6 | 213565_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5134)
SMAD6 | 213565_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0035)
SMAD6 | 213565_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD6 | 213565_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0674
p=0.2844
p=0.3099
p=0.8123
p=0.0545
p=0.5184
p=0.1197
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2